LESSON PLAN

AIRCRAFT ELECTRICAL EQUIPMENT, POWER GENERATION AND DISTRIBUTION (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	32
Total No. of Lesson Plans	35	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Direct reading magnetic compasses- Magnetic properties, hard iron and soft iron, terrestrial magnetism, magnetic variation, magnetic dip.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Direct reading magnetic compasses- Magnetic properties, hard iron and soft iron, terrestrial magnetism, magnetic variation, magnetic dip.
5. Specific Objectives	15.1.12.1 Trainee will be able to explain properties and reading of magnetic compass.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of magnetic compass theory using L & Ds and IB; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the magnetic compass in aircraft navigation?
(b) Define the difference between true north and magnetic north.
(c) List three factors that affect the accuracy of a magnetic compass.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Magnetic properties of compass needle • Hard iron and soft iron distinctions • Terrestrial magnetism fundamentals	L & Ds	PPT, IB	15 min
(b)	• Magnetic variation definition and causes • Magnetic dip and its effect on compass • Isogonic lines and agonic lines	L & Ds	PPT, PDF	20 min
(c)	• Compass deviation basics • Interpreting magnetic variation charts • Summary of magnetic properties	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how magnetic variation differs from magnetic deviation.
(b) Describe the impact of magnetic dip on the compass reading in high latitudes.
(c) Identify the difference between hard iron and soft iron in the context of compass errors.

12. Home Assignment and Follow Up	Prepare a brief note on how magnetic variation changes with location and its relevance for the next lesson on compass construction and compensation.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Features of direct reading compass- Magnetic system, liquid damping, liquid expansion, compensation, compass construction. Practical on compass, how to read compass & safety precautions.	Refer to the master syllabus reference list	LP 33

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min